INCH-POUND
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12 January 2017

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 30,000-POUND CAPACITY AT 24-INCH LOAD CENTER

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for a diesel-engine-driven, front-wheel-drive, rear-wheel-steer, pneumatic-tired, 30,000-pound (lb) capacity, forklift truck. The forklift is intended for handling cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

2. SALIENT CHARACTERISTICS.

2.1 Safety. The forklift shall comply with ANSI/ITSDF B56.1, UL 558, and Occupational Safety and Health Administration (OSHA) standards in effect at the time of manufacture. The forklift shall be OSHA DS rated. An operator's seatbelt conforming to SAE J386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip-over shall be provided.

2.2 Environmental requirements (OPTIONAL). The truck, with type A winterization system, shall be capable of operating in temperatures ranging from -65 to 125 °F. The truck, with type C winterization system, shall be capable of operating in temperatures ranging from -25 to 125 °F.

2.3 Engine. The forklift shall be powered by a standard, commercial, turbocharged diesel engine. The engine shall operate on diesel fuel in accordance with ASTM D975 without detrimental effects on the engine or its performance.

2.4 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel and biodiesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

2.5 Starting system. The starter switch shall not activate the engine starter while the engine is running, nor when the engine is not running and the transmission is in any forward or reverse gear. The starter switch shall only operate when the transmission is in the "neutral" or "park" positions.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: WR-AFLCMC/WNZEB, 235 Byron Street, Warner Robins AFB, GA 31098 or email SPEC99@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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FSC 3930

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2.6 Transmission. A torque converter and a powershift type of transmission shall be furnished with at least two forward speeds and two reverse speeds. The transmission shall provide for positive-inching control, or declutch control, that permits lifting of rated load at maximum lifting speed while the transmission control is in either the forward or reverse direction without vehicle movement. This control shall also apply the service brakes. When specified (see 6.3), a transmission oil cooler shall be furnished.

2.7 Hydraulic system. A pressure-relief protection device is required along with pump(s), cylinders, control valves, control lever, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system.

2.8 Dimensions. The overall truck height, with mast collapsed, shall not exceed 143 inches. The overall width of the truck, at the widest point, shall not exceed 102 inches. The maximum length, including the forks, shall not exceed 268 inches.

2.9 Wheelbase. The wheelbase of the truck shall not exceed 130 inches.

2.10 Uprights and carriage. With no load on the forks, the seated operator, with seatbelt engaged, shall be able to see at least one fork tip at all lift heights and fork positions. The carriage shall have a minimum width of 99 inches.

2.11 Forks and carriage. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift. Fork tines shall have the following standard dimensions.

2.11.1 Standard forks.

a. Length: 96 ±1 inches
b. Width: 8 inches
c. Thickness: 3.5 inches

2.12 Fork positioner. A hydraulically-powered fork positioner shall be furnished and controlled by the operator while seated. The fork positioner shall allow tine spacing of 3 inches (or less) to 95 inches (or more), measured between centerlines of the forks.

2.13 Side shift. Hydraulically-powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least 6-inch side shift capability (each side of center or 12 inches minimum total travel).

2.14 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical, rear-load guard of at least 48 inches high, measured from the load carrying surfaces of the forks.

2.15 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

2.16 Service brakes. Power brakes conforming to ANSI/ITSDF B56.1 shall be furnished, except maximum brake-pedal force during stopping shall not exceed 80 lbs.

2.17 Parking brake. Parking brake shall conform to ANSI/ITSDF B56.1.

2.18 Electrical system. A 12-volt, direct-current electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

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2.19 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter and tachometer shall be installed in the instrument panel. Except for the hour meter, all instruments shall illuminate. All load-motion controls shall be right-hand operation controls and of the self-centering type, i.e., controls shall return to the neutral position when released.

2.20 Lighting. A minimum of four sealed-beam floodlights shall be furnished; two in front facing forward and two mounted in the rear facing rearward. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location placement or guards. Individual operator-controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.21 Horn. A horn button shall be mounted in the center of the steering wheel.

2.22 Back-up alarm. An audible back-up alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse. The back-up alarm must conform to SAE J994.

2.23 Operator's overhead guard. The guard shall be in accordance with ANSI/ITSDF B56.1. It shall also be a falling-object protective structure (FOPS) conforming to ANSI/ITSDF B56.1. Overhead guard height from the ground to the top of guard shall be 120 inches maximum.

2.24 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate overhead protection requirements of ANSI/ITSDF B56.1. The cab shall have a door on each side of the vehicle with at least one window per door, which can be opened for ventilation; the windows shall be capable of being locked in either the open or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and roof window. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant, non-absorbent thermal insulation. The interior shall come equipped with a floor mat. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2-lb capacity, ABC type, or equivalent. The fire extinguisher shall be UL listed. The manufacturer's heaviest duty heater and defroster shall be furnished. The cab shall come equipped with an air conditioner.

2.24.1 Seat. The cab shall have a vinyl, mechanical-suspension seat with an operator presence system.

2.24.2 Handrail. The cab shall be installed with handrails to assist with operator entry and exit.

2.25 Windshield wipers. There shall be at least three wipers furnished; one on the front windshield, one on the rear window, and one on the roof window.

2.26 Mirror. The truck shall be equipped with an adjustable rearview mirror mounted within the operator’s field of view as described in SAE J985. The exact mounting location shall be determined by the manufacturer and shall increase the driver’s field of vision. The truck shall come equipped with adjustable side-view mirrors.

2.27 License plate. The truck shall come equipped with a rear, license-plate mounting bracket with plate-attaching hardware.

2.28 Performance.

2.28.1 Lifting speed. Lifting speed shall be at least 81 feet per minute without load and 71 feet per minute with rated load.

2.28.2 Lowering speed. Lowering speed shall be at least 98 feet per minute with rated load on the forks and 94 feet per minute without load.

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2.28.3 Turning radius. The maximum turning radius of the truck, measured to the outer tire, shall not exceed 180 inches.

2.28.4 Travel speed. Travel speed shall be 16 miles per hour (mph) maximum, in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that maximum speed is not exceeded for safe operations.

2.28.5 Slope ascension. The truck shall be able to accelerate up a 34 percent slope at 1 mph with rated load on the forks.

2.28.6 Upright tilt. The mast shall have at least 15 degrees of forward tilt and at least 12 degrees of rear tilt without a load on the forks.

2.28.7 Upright height (collapsed mast height) and maximum forklift height (STANDARD). With no load on the forks and the forks on the ground, the maximum upright height shall be 215 inches. The maximum forklift “lift height” shall be at least 148 inches with rated load on the forks and the forks horizontal.

2.28.8 Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 7 inches. Ground clearance at the center of the wheel base with rated load shall be a minimum of 14 inches.

2.28.9 Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 10 minutes with not more than 1.75 inches of vertical drift, and not more than 1 degree of rotational drift.

2.28.10 Stability. The forklift shall meet the ANSI/ITSDF B56.1 "forward stacking," "forward travel," "lateral stacking," and "lateral travel" stability requirements.

2.28.11 Noise limits. Maximum allowable noise level shall not exceed 84 A-weighted decibels (dBa) at the operator's ear. If the noise level at the operator's ear is between 85 to 92 dBa, a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level shall not exceed 92 dBa.

2.29 Painting. The forklift shall be primed and painted using standard commercial practices. The final paint shall be a standard, commercially-available polyurethane. The paint finish shall show no evidence of uneven application, curtains, runs, snags, orange peel, lack of adhesion, or other defects. The color shall be the manufacturer’s standard color. When specified (see 6.3), the manufacturer shall make color 24052 of FED-STD-595 available.

2.30 Markings, data plates, and instruction plates. Corrosion-resistant identification plate, hydraulic schematic, wiring schematic, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

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Nomenclature:
Make and Model:
Manufacturer's address and phone number:
Manufacturer's Serial Number (VIN):
Registration Number:
National Stock Number (NSN):
Vehicle Curb Weight (lbs):
Payload, Maximum (lbs):
Gross Weight, Maximum (lbs):
Date of Delivery:
Warranty (months/miles):
Contract Number:
Ship to _____ Mark for _____
U S Property

2.30.1 Lubrication data plate. A permanently-marked, corrosion-resistant plate directing attention to all lubrication fittings and components shall be provided. The plate shall be permanently mounted to the truck. The plate shall also identify the type and grade of lubricant required for all operational temperatures.

2.30.2 Transportation data plate. A permanently-marked, corrosion-resistant plate marked with transportation data shall be provided and mounted. The plate shall include the following information:

Side and rear silhouette views of the forklift.
Horizontal and vertical location of the center of gravity of the forklift on the silhouette view.
Shipping weight.
Loading cubage.
Overall height, width, and length.
Lift and tie-down information.

2.31 Manuals. The truck shall come equipped with an operator's, parts, and service manual. The parts manual shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. Manuals shall be provided in accordance with DFARS 252.227-7015, *Technical data-Commercial Items*.

2.32 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound, or be furnished with a nonslip metal or tread-plate surface.

2.33 Lifting and tie-down provisions. There shall be permanently-installed lifting devices that enable the forklift to be lifted in its normal travel configuration, and tie-down devices that enable the forklift to be tied down to the floor of the transportation media. All devices shall have an inside diameter of 3 inches and a minimum structural safety factor of 2:1, based on static load. The notation "LIFT HERE" and "TIEDOWN" shall be stenciled, in black, near each lifting and tie-down device.

2.34 Electromagnetic interference (EMI). The forklift shall comply with EMI requirements of SAE J551-1 and SAE J551-2.

2.35 Highway transportability. The truck, when loaded on a semitrailer, shall be within the highway permit limits for all states.

2.36 Exhaust system. The truck shall contain all of the standard commercial components composing an exhaust system. When specified (see 6.3), the exhaust system shall include a commercially available spark-arrestor muffler. The spark-arrestor muffler shall be sized to meet requirements for

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engine exhaust flow at rated engine speed.

2.37 Winterization.

2.37.1 Type A winterization. When specified (see 6.3), the truck shall be furnished with type A winterization that protects to -65 °F and shall consist of the following:

a. Power-plant heaters. Engine coolant, engine oil, and battery heaters shall be provided. All heaters shall operate on either 110 volts @ 60 Hz or 220 volts @ 50 Hz alternating current (VAC). A three-wire, 25-feet long, weatherproof cable of adequate capacity for all heaters being used simultaneously shall be provided. The cable shall be able to connect to NATO and American connectors. A stowage place shall be provided on the truck to keep the cable in when not in use. The heaters shall be as follows:

(1) The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant-circulating pump, driven by a 110 or 220 VAC motor shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of 10 °F in an ambient temperature of -65 °F. It shall be controlled to limit engine coolant to not more than 150 °F.

(2) An engine oil heater with adequate capacity to maintain engine oil at a temperature of at least 10 °F in an ambient temperature of -65 °F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 °F.

(3) A battery heater shall be provided. It shall have adequate capacity to maintain battery electrolyte at a temperature of at least 10 °F in an ambient temperature of -65 °F, and shall have a thermostat to limit temperature of the electrolyte to not more than 80 °F.

b. Cab. The cab shall meet the requirements of 2.24, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.37.1.c.

c. Cab heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of 40 °F at cab floor level in an ambient temperature of -40 °F.

d. Starting aid. Glow plug(s) or a measured-shot, ether injection system shall be provided to assist in engine cold-temperature starting.

e. Antifreeze. Engine coolant shall be protected to -65 °F with antifreeze.

2.37.2 Type C winterization. When specified (see 6.3), the truck shall be furnished with type C winterization that protects to -25 °F and shall consist of the following:

a. Starting aid. Glow plug(s) or a measure-shot, ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements of 2.24, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.37.2.c.

c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least 40 °F at cab floor level in an ambient temperature of -20 °F.

d. Antifreeze. Engine coolant shall be protected to -25 °F with antifreeze.

3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials or biobased content to the maximum extent practicable, in accordance with Federal Acquisition Regulation, part 23.403.

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4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING. Unless otherwise specified (see 6.3), the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Intended use. This CID covers the general requirements for a diesel-engine-driven, front-wheel-drive, rear-wheel-steer, pneumatic-tired, 30,000-lb capacity, forklift truck. The forklift is intended for handling cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

6.2 Source of documents.

6.2.1 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://assist.dla.mil.

6.2.2 ANSI/ITSDF standards. Copies of ANSI/ITSDF standards may be obtained from the Industrial Truck Standards Development Foundation, Suite 460, 1750 K Street NW, Washington, DC 20006, or online at http://www.itsdf.org/default.asp.

6.2.3 SAE standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org/.

6.2.4 TRA standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Monroe West Avenue, Suite 150, Copley, OH 44321, or online at http://www.us-tra.org/publications.html.

6.2.5 UL standards. Copies of UL standards may be obtained from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062, or online at http://www.ul.com/.

6.3 Ordering data. Acquisition documents shall specify the following information:

a. CID document number, revision, and CID PIN.

b. Transmission oil cooler, when specified (see 2.6).

c. When paint is other than manufacturer's standard color (see 2.29).

d. Spark-arrestor muffler, if required (see 2.36).

e. Type of winterization, if any (see 2.37.1 or 2.37.2).

f. Packaging requirements, if different (see 5).

6.4 Subject term (key word) listing.

Front-wheel drive
Rear-wheel steer

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Handling cargo
Warehouses
Fork positioner
Load backrest
Cab

MILITARY INTERESTS

Custodian:
Air Force - 184

Preparing Activity:
Air Force - 184

Agent:
Air Force - 99

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